Fiscal Year:	FY 2012	**Task Last Updated:**	FY 03/17/2013
PI Name:	Muday, Gloria Ph.D.		
Project Title:	Dynamics of auxin transport protein localization and gravitropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	08/01/2009	**End Date:**	01/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Extended to 1/31/2014, per NSSC information (Ed., 7/25/13) NOTE: Extended to 7/31/2013, per NSSC information (Ed., 12/18/2012) NOTE: extended to 12/31/2012 per NSSC information (Ed., 4/23/2012)		
Key Personnel Changes/Previous PI:			
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Task Description:	Plants respond to gravity in numerous ways; the best understood is the change in growth orientation in response to reorientation of plants relative to the gravity vector. A growth asymmetry develops, which allows the plant organ to return to the original angle of growth relative to gravity. Understanding gravitropism is essential to facilitate growth of plants in the microgravity environment of space. Gravitropic bending is mediated, at least in part, by redistribution of the plant hormone auxin. This proposal examines the mechanisms by which the cellular asymmetry of auxin transport proteins is established and the molecular mechanisms that allow translocation of the auxin signals from the site of perception to the site of reaction. We will employ three approaches to transiently alter the expression and localization of auxin transport proteins to examine the effect on gravitropic curvature in roots of the model species Arabidopsis thaliana. First, we will use a temperature sensitive mutant, scd1-1, with a defect in endomembrane targeting, to transiently disrupt, auxin transport protein localization and gravitropic curvature. Second, we will transiently perturb the activity and/or localization of auxin transport proteins through a pharmacological approach and measure the resulting effect on gravitropic kinetics. In our third approach, we will use well characterized null mutants with defects in auxin transport proteins and will express wild type copies of defective genes in an inducible expression system. This approach of reintroducing a gene function at a controlled time points during the gravitropic process, should allow us to understand when these proteins function in controlling auxin transport and gravitropism. The planned experiments will combine high resolution laser scanning confocal microscopy and high temporal resolution gravitropic analyses with transient perturbation of auxin transport to understand the dynamics of auxin transport protein localization and resulting auxin asymmetries that drive differential gravitropic growth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During this year of funding (June 1, 2011-May 31, 2012) we completed our analysis of the gravitropic and auxin transport phenotypes of the temperature sensitive scd1-1 mutant, which has a defect in a gene encoding a protein predicted to function in Rab GTPase-dependent protein trafficking. All aspects of root growth and gravitropism are normal in this mutant at the permissive temperature of 18°C. At the nonpermissive temperature of 25°C, scd1-1 exhibits significant reductions in gravitropic curvature. We monitored gravitropic curvature, growth rate, and auxin transport in scd1-1 as seedlings were switched from growth at the permissive temperature (18°C) to the non-permissive temperature (25°C). After a 90° reorientation relative to the gravity vector, the wild type plants grown at 25°C show a gravitropic response is characterized by a 90° root angle within 8 hours, while scd1-1 has delayed gravitropic response reaching only a 45° root angle after 8 hours. To determine the rate of response to a change in temperature in the mutant, gravitropic curvature after transition of seedlings from the permissive to nonpermissive temperature was analyzed. The reductions in root gravitropic curvature in scd1-1 are evident in seedlings that were exposed to the nonpermissive temperature (25°C) for 8 or 24 hours. Second, we examined the transport of auxin in the scd1-1 mutant, to determine if the mechanism by which gravity response is impaired is tied to the ability to transport IAA. Basipetal or shootward IAA transport is reduced in the scd1-1 mutant relative to wild-type at 25°C, while similar transport amounts are detected at 18°C. These differences were observed using both traditional radioactive IAA transport assay and by applying auxin at the root tip and measurement of the shootward progression of fluorescent signal resulting from auxin induced gene expression, using the DR5:GFP reporter. Just 8 hours of incubation at RT (25°C) after growth at 18°C is sufficient to show a significant reduction in IAA transport, nicely mirroring the timing of the defect in gravitropism. The third line of experiments in scd1-1 focuses on examination of the effects of temperature changes on the distribution of auxin transport proteins in this mutant background. We have used laser scanning confocal microscopy to examine the tissue level and sub-cellular distribution of auxin efflux carriers PIN1, PIN2, PIN3, AUX1, and ABCB19 that are translational fusions to GFP, with PIN2:GFP being best linked to gravity response. The pattern and intensity of fluorescence was similar in wild-type and scd1-1 at 18°C. At 25°C, the PIN1 and PIN2:GFP fusion in scd1-1 were found at substantially lower levels in the plasma membrane, and accumulated in endomembranes, while changes in the other reporters were subtle. The membrane asymmetries of PIN1:GFP and PIN2:GFP reporters were generally maintained, although substantially weaker. The apparent structure and character of endomembranes that accumulated PIN1-GFP and PIN2-GFP were quite different. This result suggests that in the mutant, these two auxin transport proteins may be trafficked by different pathways in the different cell types in which they are found, causing them to be shunted to different structures where they accumulate. In contrast, PIN3-GFP looked qualitatively the same in both wild type and scd1-1 at the subcellular level, but was restricted to fewer cells at 25°C compared to wild type. During this year, an undergraduate student focused on examining the kinetics of this response, doing live imaging of PIN1:GFP labeled endomembrane organelles and Flourescence Recovery After Photobleaching (FRAP), which quantifies the rate of return of fluorescently labeled protein to a location on the membrane that is bleached by a laser. These experiments demonstrated that PIN1:GFP fluorescence returns to the plasma membrane at a slower rate in the scd1-1 mutant than in wild-type, consistent with a targeting defect in the scd1-1 mutant. Together, these results are a complete story linking SCD1 protein to control of targeting of auxin transport proteins to the appropriate location in the cell to control auxin transport and gravitropism. The final step of these experiments is identification of the nature of the endomembranes that accumulate PIN2:GFP. We have crossed endomembrane specific fluorescent markers for early and late endosomes, Golgi, endoplasmic reticulum, and vacuole into scd1-1xPIN1 GFP and scd1-1xPIN2GFP. During this year of funding we have begun imaging of these lines with multiple reporters. If an endomembrane specific marker line shows fluorescence that colocalizes with the observed PIN: GFP endomembranes, this will provide insight into the role of SCD1 protein in targeting auxin transport proteins, which are then needed for gravitropic responses. We have received a no-cost extension to complete these imaging experiments in a final year of funding. A second line of experimentation is utilizing a pharmacological approach to transiently disrupt activity and/or targeting of auxin transport proteins to understand their role in gravitropic responses. We have examined compounds which inhibit protein phosphorylation and dephosphorylation, specifically staurosporine and cantharidin, respectively, and combined these inhibitors with mutants in specific kinase and phosphatase genes, to provide evidence of reciprocal regulation of auxin transport by protein phosphorylation and regulation of gravitropic responses (Sukumar et al., 2009). During this year of funding another undergraduate student completed a study of three endosidin compounds, ES10,

	ES11, and ES12, on gravity response and auxin transport. These endosidin chemicals were isolated in a chemical genomic screen by Glenn Hicks and Natasha Raikhel at UC Riverside for inhibition of hypocotyl gravitropism. These experiments quantified the kinetics of inhibition of root gravitropism by these three compounds using high resolution imaging and identified doses of compounds that inhibited gravitropism, but not root elongation. ES11 and ES12, but not ES10, perturb the formation of asymmetric gradients of auxin induced gene expression across a gravity stimulated root. The effect of these compounds on auxin transport was quantified and only ES12 reduced shootward or basipetal IAA polar transport, which supplies auxin for root gravitropic curvature. Therefore these three compounds fall into three distinct activity pathways, suggesting they act on different protein targets. Finally, the effect of these compounds on gravity response in mutants that have altered gravitropic responses are still underway, but initial experiments suggest that agravitropic mutants show differential responses to these drugs. These mutant experiments are also to be completed during our no cost extension as is the publication describing these results.
Bibliography Type:	Description: (Last Updated: 08/25/2020)
Abstracts for Journals and Proceedings	Falbel T, Isley J, Mattoz C, Muday G. "Altered Auxin Transport and Gravitropic Response in the scd1 Mutant." Plant Biology 2011 (American Society of Plant Biologists), Minneapolis, MN, August 6-10, 2011. Plant Biology 2011 (American Society of Plant Biologists), Minneapolis, MN, August 6-10, 2011. Abstract # P16053. http://abstracts.aspb.org/pb2011/public/P16/P16053.html , Aug-2011
Dissertations and Theses	Resayat A. "Understanding the role of SCD1 in targeting PIN1 protein in Arabidopsis thaliana." Undergraduate Honors Thesis, Wake Forest University, May 2012. , May-2012